Linear Models Lecture 14: 2SLS, LATE, and the MTE Framework

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Control Function Regression

- Structural model with endogenous X_2:

$$Y = \boldsymbol{x}_1'\beta_1 + \boldsymbol{x}_2'\beta_2 + e$$

$$\boldsymbol{x}_2 = \Gamma_{12}'\boldsymbol{z}_1 + \Gamma_{22}'\boldsymbol{z}_2 + u_2$$

- The **control function** approach directly models the dependence between e and u_2:

$$e = u_2'\alpha + v, \qquad \alpha = (E[u_2 u_2'])^{-1}E[u_2 e], \qquad E[u_2 v] = 0$$

- Substituting into the structural equation:

$$Y = X_1'\beta_1 + X_2'\beta_2 + u_2'\alpha + v$$

- After controlling for u_2: $E[X_1 v] = E[X_2 v] = E[u_2 v] = 0$.

Control Function: Implementation

- **Step 1:** Estimate the first stage by OLS to get residuals:

$$\hat{u}_{2i} = X_{2i} - \hat{\Gamma}'_{12} Z_{1i} - \hat{\Gamma}'_{22} Z_{2i}$$

- **Step 2:** Include $\hat{u}_2$ as an additional regressor:

$$Y = X\hat{\beta} + \hat{U}_2 \hat{\alpha} + \hat{v}$$

- This is "subtracting off the endogenous part" of X_2.
- Under homoskedasticity, the control function estimator of β is numerically identical to 2SLS.
- The coefficient $\hat{\alpha}$ directly tests endogeneity — if $\alpha = 0$, then X_2 is exogenous.

R: Control Function

```r
# Step 1: First stage
first_stage <- lm(X2 ~ Z1 + Z2, data = dat)
dat$u2_hat <- residuals(first_stage)

# Step 2: Structural equation with control
cf_reg <- lm(Y ~ X1 + X2 + u2_hat, data = dat)
summary(cf_reg)
# coef on u2_hat tests endogeneity (t-test on alpha)
# coef on X2 is the IV estimate of beta_2
```

The control function makes identification transparent: we "control for" the endogenous component of X_2. The t-test on $\hat{\alpha}$ is equivalent to the Hausman test.

IV Has No Finite Moments

- A striking result (Kinal 1980, Kiviet): the IV estimator has at most $q = \ell - k$ moments, where q is the number of overidentifying restrictions.
- When just identified ($q = 0$): IV has no expectation and no variance.
- Why? Consider $\hat{\beta}_{IV} = (Z'X)^{-1}Z'Y$. The denominator $Z'X$ can be arbitrarily close to zero, creating Cauchy-like tails.
- Consequences:
  - The **bootstrap fails** in the just-identified case — it requires finite variance.
  - "Bias" is not well-defined in the usual sense — we work with approximate bias instead.

Imprecision of IV

- Suppose Z and X are scalar, mean zero. Compare asymptotic variances:

$$Avar(\hat{\beta}_{OLS}) = \frac{\sigma_e^2}{n} \cdot \frac{1}{\text{var}(x)}$$

$$Avar(\hat{\beta}_{IV}) = \frac{\sigma_e^2}{n} \cdot \frac{1}{\text{var}(x)} \cdot \frac{1}{\rho_{xz}^2}$$

- Therefore:

$$\boxed{Avar(\hat{\beta}_{IV}) = Avar(\hat{\beta}_{OLS}) \cdot \frac{1}{\rho_{xz}^2}}$$

- As $\rho_{xz}^2 \to 0$: $Avar(\hat{\beta}_{IV}) \to \infty$.
- IV is **always** less precise than OLS — the cost of correcting for endogeneity.

Decomposing the IV Estimation Error

- Write: $y = X\beta + e$ and $X = Z\pi + v$. Then:

$$\begin{aligned}
\hat{\beta}_{IV} &= (X'P_Z X)^{-1} X'P_Z y \\
&= \beta + (X'P_Z X)^{-1} X'P_Z e \\
&= \beta + (X'P_Z X)^{-1} (\pi'Z' + v')P_Z e \\
&= \beta + \underbrace{(X'P_Z X)^{-1} \pi'Z'e}_{\text{Term A}} + \underbrace{(X'P_Z X)^{-1} v'P_Z e}_{\text{Term B}}
\end{aligned}$$

- **Term A:** Involves $Z'e$. Since $E[Z'e] = 0$, this vanishes in expectation.
- **Term B:** Involves $v'P_Z e$. Since v and e are correlated (endogeneity!), this does **not** vanish.
- Term B is the source of finite-sample bias.

Approximate Bias of IV

Taking approximate expectations (since exact ones may not exist):

$$\begin{aligned} E(\hat{\beta}_{IV}) - \beta &\approx (E(X'P_Z X))^{-1} E(v'P_Z e) \\ &= (E[(\pi' Z' + v')P_Z(Z\pi + v)])^{-1} E[v'P_Z e] \\ &= (E[\pi' Z' Z\pi] + E[v'P_Z v])^{-1} E[v'P_Z e] \end{aligned}$$

where the cross terms vanish because $E[Z'v] = 0$.
Using $E[v'P_Z v] = \sigma_v^2 \cdot p$ and $E[v'P_Z e] = \sigma_{ev} \cdot p$ (where $p = \ell$ is the number of instruments):

$$E(\hat{\beta}_{IV}) - \beta \approx (E[\pi' Z' Z\pi] + \sigma_v^2 p)^{-1} \sigma_{ev} \cdot p$$

Bias in Terms of the F-Statistic

Dividing numerator and denominator by $\sigma_v^2 p$:

$$E(\hat{\beta}_{IV}) - \beta \approx \frac{1}{\left(\frac{E[\pi' Z' Z \pi]/p}{\sigma_v^2} + 1\right)} \cdot \frac{\sigma_{ev}}{\sigma_v^2} \approx \frac{1}{1 + F_{p,n-p}} \cdot \frac{\sigma_{ev}}{\sigma_v^2}$$

- F is the first-stage F-statistic testing $H_0 : \pi = 0$.
- The term $\frac{\sigma_{ev}}{\sigma_v^2}$ is exactly the **OLS bias** (endogeneity).
- Key implications:
  - As $F \to \infty$: bias $\to 0$ (strong instruments eliminate bias)
  - As $F \to 0$: bias $\to \frac{\sigma_{ev}}{\sigma_v^2}$ = OLS bias (IV provides no correction)
  - Adding irrelevant instruments increases p, decreases F, increases bias

Rule of Thumb and Modern Diagnostics

- **Staiger & Stock (1997):** Rule of thumb $F > 10$ for reliable IV.
- **Stock & Yogo (2005):** Critical values for the F-statistic that ensure IV bias is at most x% of OLS bias.
- **Lee et al. (2022):** The tF procedure — adjust critical values for the t-test based on the first-stage F.
- The bias formula explains why:
  - "Fishing" for instruments is dangerous — more instruments $\Rightarrow$ more bias
  - A single strong instrument is often better than many weak ones
  - The just-identified case ($p = k_2$) minimizes the finite-sample bias problem

Weak Instruments: OLS Asymptotic Bias

Scalar case: single x, single instrument z. An instrument is **weak** if ρ_{zx} is small.

$$
\begin{aligned}
\text{plim } \hat{\beta}_{OLS} &= \text{plim } \frac{\text{cov}(x, y)}{\text{var}(x)} = \text{plim } \frac{\text{cov}(x, x\beta + e)}{\text{var}(x)} \\
&= \beta + \text{plim } \frac{\text{cov}(x, e)}{\text{var}(x)} \\
&= \beta + \frac{\text{cov}(x, e)}{\sqrt{\text{var}(x)}\sqrt{\text{var}(e)}} \cdot \frac{\sqrt{\text{var}(e)}}{\sqrt{\text{var}(x)}} \\
&= \beta + \rho_{xe} \frac{\sigma_e}{\sigma_x}
\end{aligned}
$$

The OLS asymptotic bias is $\rho_{xe} \frac{\sigma_e}{\sigma_x}$.

Weak Instruments: IV Asymptotic Bias

$$
\begin{aligned}
\text{plim } \hat{\beta}_{IV} &= \text{plim } \frac{\text{cov}(z,y)}{\text{cov}(z,x)} = \beta + \text{plim } \frac{\text{cov}(z,e)}{\text{cov}(z,x)} \\
&= \beta + \frac{\rho_{ze}}{\rho_{zx}} \cdot \frac{\sigma_e}{\sigma_x} \\
&= \beta + \frac{\rho_{ze}}{\rho_{zx}\rho_{xe}} \cdot \underbrace{\rho_{xe}\frac{\sigma_e}{\sigma_x}}_{ABias(\hat{\beta}_{OLS})}
\end{aligned}
$$

- If the instrument is perfectly valid ($\rho_{ze} = 0$): IV is consistent regardless of ρ_{zx}.
- If the instrument is even slightly invalid ($\rho_{ze} \neq 0$): weak relevance ($\rho_{zx} \approx 0$) amplifies the bias.

When IV Is Worse Than OLS

$$\frac{ABias(\hat{\beta}_{IV})}{ABias(\hat{\beta}_{OLS})} = \frac{\rho_{ze}}{\rho_{zx}\rho_{xe}}$$

If $\frac{\rho_{ze}}{\rho_{zx}\rho_{xe}} \geq 1$, then IV is more biased than OLS.

- **Example:** $\rho_{xe} = 0.5$ (very endogenous X), $\rho_{ze} = 0.01$ (barely invalid Z), $\rho_{zx} = 0.019$ (weak instrument):

$$\frac{0.01}{0.019 \times 0.5} = 1.052 \quad \Rightarrow \quad \text{IV bias exceeds OLS bias}$$

- Approximate relationship to the F-statistic:

$$\frac{ABias(\hat{\beta}_{IV})}{ABias(\hat{\beta}_{OLS})} \approx \frac{1}{F}$$

An F of 100 means IV is ∼1% as biased as OLS. An F of 5 means ∼20%.

Testing Endogeneity: Durbin-Wu-Hausman Test

- If X is exogenous, both OLS and IV are consistent, but OLS is efficient (BLUE).
- If X is endogenous, only IV is consistent.
- The **Hausman test** compares the two:

$$H = (\hat{\beta}_{IV} - \hat{\beta}_{OLS})'[Avar(\hat{\beta}_{IV}) - Avar(\hat{\beta}_{OLS})]^{-1}(\hat{\beta}_{IV} - \hat{\beta}_{OLS}) \sim \chi^2_{k_2}$$

- Rejecting H_0: either X is endogenous **or** Z is an invalid instrument.
- Equivalent to testing $\hat{\alpha} = 0$ in the control function regression.

DWH $\to$ GMM Distance Form

- The classical Hausman statistic

$$H = (\hat{\beta}_{IV} - \hat{\beta}_{OLS})' \left[\text{Avar}(\hat{\beta}_{IV}) - \text{Avar}(\hat{\beta}_{OLS})\right]^{-1} (\hat{\beta}_{IV} - \hat{\beta}_{OLS})$$

requires the difference of variances to be positive semidefinite — which holds only under conditional homoskedasticity. Under heteroskedasticity the inverse can fail.

- **Fix (Lecture 16):** the GMM *distance test*. Compute efficient GMM twice — $\bar{J}$ with the original instruments, $\hat{J}$ adding X_2 as its own instrument (valid under $H_0 : E[X_2 e] = 0$):

$$C = \hat{J} - \bar{J} \xrightarrow{d} \chi^2_{k_2} \quad \text{under } H_0.$$

- Same null as DWH; no homoskedasticity required. Report this under heteroskedasticity.

Testing Overidentifying Restrictions: Sargan Test

- When $\ell > k$, we have $q = \ell - k$ "extra" moment conditions.
- If the model is correct, all moment conditions should be approximately satisfied.
- The **Sargan test** (also called Hansen's J-test):

$$J = n \cdot \hat{e}' P_Z \hat{e} / \hat{\sigma}^2 \sim \chi^2_q \quad \text{under } H_0$$

  where $\hat{e} = Y - X\hat{\beta}_{2SLS}$.
- Rejecting H_0: at least one instrument is invalid (correlated with e).
- Limitation: If **all** instruments are invalid in the same way, the test has no power.

GMM connection: The Sargan/J-test is the overidentification test of GMM.

R: IV Estimation and Diagnostics

```
library(estimatr); library(lmtest)

# 2SLS with robust SEs
iv_fit <- iv_robust(Y ~ X2 + X1 | Z2 + X1, data = dat)
summary(iv_fit)  # coefs, robust SEs, first-stage F

# First-stage F-test (check instrument strength)
first <- lm(X2 ~ Z2 + X1, data = dat)
linearHypothesis(first, "Z2 = 0")

# OLS for comparison
ols_fit <- lm(Y ~ X2 + X1, data = dat)
```

What Does IV Estimate Under Heterogeneity?

- So far: IV estimates the “structural parameter” β.
- But what if treatment effects are **heterogeneous**? β_i varies across individuals.
- Angrist & Imbens (1994): with a binary instrument, IV estimates the **Local Average Treatment Effect**:

$$\hat{\beta}_{IV} \xrightarrow{p} \text{LATE} = E[Y_i(1) - Y_i(0) \mid \text{Compliers}]$$

- “Compliers” $=$ individuals whose treatment status is changed by the instrument.

Compliance Types

Principal Strata

Always-Takers
$D(0) = 1, \; D(1) = 1$ → Not affected by Z

Compliers
$D(0) = 0, \; D(1) = 1$ → **LATE identified here**

Never-Takers
$D(0) = 0, \; D(1) = 0$ → Not affected by Z

Defiers (ruled out)
$D(0) = 1, \; D(1) = 0$ → Excluded by monotonicity

LATE Assumptions (Angrist-Imbens)

Four assumptions for LATE identification:

1. **Independence:** $(Y_i(0), Y_i(1), D_i(0), D_i(1)) \perp Z_i$
2. **Exclusion:** Z_i affects Y_i only through D_i
3. **Monotonicity:** $D_i(1) \geq D_i(0)$ for all i (no defiers)
4. **First stage:** $E[D_i \mid Z_i = 1] \neq E[D_i \mid Z_i = 0]$

Under these assumptions, the Wald estimator identifies:

$$\frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]} = E[Y_i(1) - Y_i(0) \mid \text{Compliers}]$$

LATE is **instrument-dependent**: different instruments $\Rightarrow$ different compliers $\Rightarrow$ different LATEs.

Example: Returns to Education

- Model: $\text{lwage}_i = \alpha + \gamma_1 \text{educ}_i + v_i \cdot \text{educ}_i + u_i$
- Heterogeneous returns: $\beta_i = \gamma_1 + v_i$
- If we instrument education with quarter of birth (Angrist & Krueger 1991):
  - Compliers = those who stay in school because of compulsory schooling laws
  - LATE = return to education for **marginal students** (those induced to stay)
  - This may differ from ATE or ATT
- **Policy relevance:** LATE answers "what is the return for people affected by the policy?" — exactly the right parameter for evaluating that policy.
- Caveat: AK control for state and year-of-birth fixed effects. Whether the LATE interpretation survives the inclusion of those controls is exactly the question Blandhol et al. (2025) raise — see next slide.

When Is TSLS *Actually* LATE? (Blandhol et al. 2025)

- The LATE interpretation of TSLS hides a requirement: rich covariates,

$$\mathbb{E}[Z \mid X] = \mathbb{L}[Z \mid X].$$

  The conditional mean of Z given X must coincide with its linear projection.
- Holds automatically when there are no covariates (the Wald case), the spec is *saturated* in X, or $Z \perp X$ by design.
- Otherwise the TSLS estimand mixes complier effects with always-taker effects, and *some always-taker weights are necessarily negative* — it is not even a non-negatively-weighted average of subgroup treatment effects.
- In a survey of 99 published IV-with-covariates papers, BBMT find *one* that used a saturated specification.

TSLS-LATE: Practical Workflow (BBMT 2025)

1. **Report TSLS without covariates** if exogeneity doesn't require them. Avoid kitchen-sink controls — adding covariates makes rich covariates *less* likely to hold.
2. **Report the Ramsey RESET test** for a regression of Z on X (R: `lmtest::resettest`; Stata: `estat ovtest`). The null is rich covariates; rejection means the LATE interpretation fails.
3. **If RESET rejects, estimate β_{rich} via DDML PLIV** and report alongside TSLS. R packages: `ddml`, `DoubleML`.
4. **If Z is binary, also report the unconditional ACR/LATE.** Equals the unconditional LATE when treatment is binary; estimated via DDML or IPSW (Słoczyński et al. 2024).

The MTE framework (next section) is the principled alternative when you want to know which causal parameter you are estimating.

The Generalized Roy Model

- Potential outcomes: $Y_i(1)$, $Y_i(0)$ with heterogeneous gains $\Delta_i = Y_i(1) - Y_i(0)$.
- The treatment decision is based on **net utility**:

$$D_i^* = \mu_D(Z_i) - U_{Di}$$

$$D_i = \mathbb{1}[D_i^* > 0] = \mathbb{1}[\mu_D(Z_i) > U_{Di}]$$

- $\mu_D(Z_i)$: observable component of the treatment decision (driven by instruments).
- U_{Di}: unobservable **resistance to treatment** — the utility cost of participating.
  - Low U_{Di}: individual has low cost $\Rightarrow$ eager to participate.
  - High U_{Di}: individual has high cost $\Rightarrow$ reluctant to participate.
- Normalize $U_D \sim U[0, 1]$ (via probability integral transform).

Utility and Selection

- In the Roy framework, agents choose treatment when net utility is positive:

$$D_i = 1 \quad \Leftrightarrow \quad \underbrace{\text{Expected benefit of treatment}}_{\text{depends on } \Delta_i} > \underbrace{\text{Cost of treatment}}_{U_{Di}}$$

- The propensity score $P(Z_i) = \Pr(D_i = 1 \mid Z_i) = \mu_D(Z_i)$ summarizes the instrument.
- An individual participates iff $U_{Di} \leq P(Z_i)$.
- **Key insight:** U_D orders individuals from most to least eager.
    - $U_D \approx 0$: would participate under almost any instrument value ("always-takers")
    - $U_D \approx 1$: would almost never participate ("never-takers")
    - $U_D \in [P(z_0), P(z_1)]$: participate when $Z = z_1$ but not $Z = z_0$ ("compliers")

The Marginal Treatment Effect (Heckman & Vytlacil 2005)

Definition (Marginal Treatment Effect)

$$\Delta^{MTE}(x, u_D) = E[Y(1) - Y(0) \mid X = x, U_D = u_D]$$

- MTE is the treatment effect for individuals **at the margin** — those who are just indifferent between treatment and control when their unobserved resistance equals u_D.
- As u_D increases from 0 to 1, we trace out effects from the most eager to the most reluctant:
  - Low u_D: low-cost individuals (high utility from treatment) — often highest returns.
  - High u_D: high-cost individuals — often lowest returns (if gains and costs are correlated).
- MTE is typically **downward-sloping**: those who self-select into treatment tend to benefit most (essential heterogeneity).

All Treatment Parameters as Weighted MTE

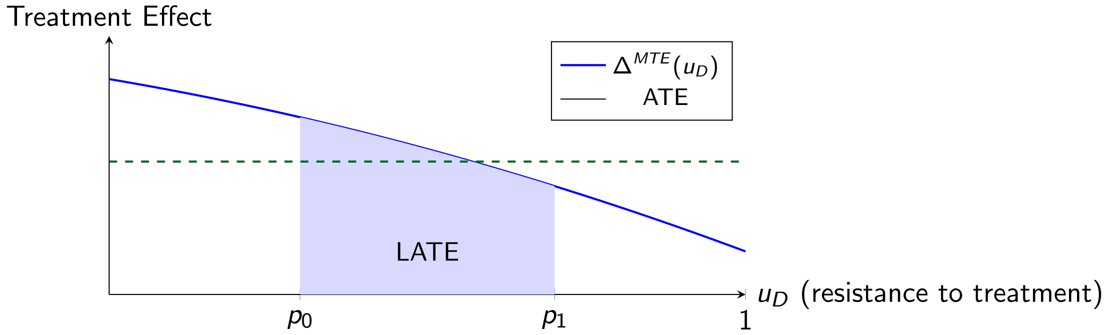
- Heckman & Vytlacil's key insight: **every** treatment parameter is a weighted average of MTE:

$$\Delta^j(x) = \int_0^1 \Delta^{MTE}(x, u_D)\,\omega_j(x, u_D)\,du_D$$

Parameter	**Weight** ω_j	**Who?**
ATE	1	Everyone
ATT	$\frac{u_D}{E[D\|X]}$ (overweights low u_D)	Treated
ATU	$\frac{1-u_D}{E[1-D\|X]}$ (overweights high u_D)	Untreated
LATE	$\frac{1}{u_D'-u_D}\mathbb{1}[u_D \leq t \leq u_D']$	Compliers

- LATE is MTE averaged over a **specific interval** of u_D values.

Visualizing the MTE Curve



LATE averages MTE over complier interval $[p_0, p_1]$. ATE averages the entire curve.

What Does Conventional IV Recover?

- With a continuous instrument, 2SLS recovers a **variance-weighted average** of LATEs:

$$\beta_{IV} = \int_0^1 \Delta^{MTE}(u_D)\, \omega_{IV}(u_D)\, du_D$$

  where ω_{IV} weights by the "first-stage intensity" at each margin.
- **Key points** (Hull 2024):
  - Under monotonicity + exclusion: IV estimates a positively-weighted average of individual effects
  - Without monotonicity: weights can be negative (hard to interpret)
  - The "model" matters for extrapolation beyond the complier population

Policy Relevance

- Different policies target different margins of the MTE curve.
- The **Policy-Relevant Treatment Effect** (PRTE):

$$\Delta^{PRTE} = \int_0^1 \Delta^{MTE}(u_D)\,\omega_{PRTE}(u_D)\,du_D$$

  where ω_{PRTE} depends on how the policy shifts participation.
- LATE from one instrument is generally **not** the right parameter for a different policy.
- Estimating the full MTE curve (under stronger assumptions) allows us to evaluate **any** proposed policy.

The MTE framework unifies the structural and treatment-effects traditions: it shows exactly what each estimand identifies and what assumptions are needed.

R: Estimating MTE with ivmte

Angrist & Evans (1998): effect of fertility on labor supply. Y = worked, D = morekids, Z = samesex.

```
library(ivmte)  # Mogstad, Santos, Torgovitsky (2018)

# ATT via MTE extrapolation
att <- ivmte(data = AE, target = "att",
  m0 = ~ u + yob, m1 = ~ u + yob,
  ivlike = worked ~ morekids + samesex,
  propensity = morekids ~ samesex + yob)

# ATE -- just change target
ate <- ivmte(data = AE, target = "ate",
  m0 = ~ u + yob, m1 = ~ u + yob,
  ivlike = worked ~ morekids + samesex,
  propensity = morekids ~ samesex + yob)
```

R: ivmte — LATE and Richer Specifications

```
# LATE for specific complier group
late <- ivmte(data = AE, target = "late",
  late.from = c(samesex = 0),
  late.to   = c(samesex = 1),
  m0 = ~ u + I(u^2) + yob,
  m1 = ~ u + I(u^2) + yob,
  ivlike = worked ~ morekids + samesex,
  propensity = morekids ~ samesex + yob)
```

- `m0`, `m1`: marginal treatment response functions; `u` is the unobserved resistance U_D.
- `target`: `"ate"`, `"att"`, `"atu"`, `"late"`, or `"genlate"`.
- Polynomial terms in `u` allow flexible MTE curves.
- The package implements Mogstad, Santos & Torgovitsky (2018, *Econometrica*) — bounds when point identification fails.

Why We Need GMM: The MTE Estimation Problem

- The MTE framework poses an estimation challenge that 2SLS alone cannot solve.
- In `ivmte`, the `ivlike` argument specifies **multiple IV-like moment conditions**:

$$E[Z_i(Y_i - m_0(\boldsymbol{x}_i, U_{Di}; \theta)(1 - D_i) - m_1(\boldsymbol{x}_i, U_{Di}; \theta)D_i)] = 0$$

- We want to recover θ (the MTR parameters) from these moments, but:
  1. We may have **more moments than parameters** (overidentification)
  2. The model is **nonlinear** in θ (MTR functions can be flexible polynomials in u_D)
  3. Under heteroskedasticity, different moments have different precision
- 2SLS handles (1) but only for linear models. We need a **general framework** for combining moment conditions optimally.

From IV Moments to GMM

- Recall: 2SLS solves the overidentified moment conditions

$$E[Z_i(Y_i - X_i'\beta)] = 0$$

  using the weighting matrix $(Z'Z/n)^{-1}$.
- The **Generalized Method of Moments** generalizes this:
  1. **Efficiency:** Under heteroskedasticity, 2SLS is not efficient. GMM uses the optimal weighting $\hat{W} = \hat{\Omega}^{-1}$.
  2. **Generality:** GMM applies to **any** moment conditions $E[g(W_i, \theta)] = 0$ — including the nonlinear moments from MTE estimation.
  3. **Testing:** The GMM J-statistic generalizes the Sargan test.
- MTE estimation via `ivmte` is a GMM problem: find MTR parameters that best fit the IV-like moments, subject to shape constraints.

Preview: The GMM Estimator

- Given moment conditions $E[g(W_i, \theta_0)] = 0$ with $\dim(g) > \dim(\theta)$:

$$\hat{\theta}_{GMM} = \arg\min_{\theta} \left[\frac{1}{n}\sum_{i=1}^{n} g(W_i, \theta)\right]' \hat{W} \left[\frac{1}{n}\sum_{i=1}^{n} g(W_i, \theta)\right]$$

- Special cases:
  - Linear IV moments + $\hat{W} = (Z'Z/n)^{-1}$: **2SLS**
  - Linear IV moments + $\hat{W} = \hat{\Omega}^{-1}$: **efficient IV-GMM**
  - MTR moments + shape constraints: **MTE estimation** (Mogstad, Santos & Torgovitsky 2018)
- The *J*-test: $J = n \cdot \bar{g}(\hat{\theta})' \hat{W} \bar{g}(\hat{\theta}) \sim \chi^2_q$ tests overidentifying restrictions.

Next week: Full development of GMM — estimation, inference, and applications.

Summary

1. The **control function** makes endogeneity correction explicit by adding $\hat{u}_2$ as a regressor.
2. IV has no finite moments when just identified — bootstrap fails.
3. IV is always less precise than OLS: $Avar(\hat{\beta}_{IV}) = Avar(\hat{\beta}_{OLS})/\rho^2_{xz}$.
4. IV bias $\approx$ OLS bias $\times \frac{1}{1+F}$; weak instruments ($F < 10$) are dangerous.
5. The **Hausman test** detects endogeneity; **Sargan test** detects invalid instruments.
6. Under heterogeneous effects, IV estimates **LATE** — the effect for compliers.
7. The **MTE framework**: all treatment parameters are weighted averages of $\Delta^{MTE}(u_D)$, indexed by unobserved resistance (utility cost).
8. 2SLS is a special case of **GMM** with a specific weighting matrix.